

Acid-Base Titration and pH Measurement

Objective

The objective of this laboratory exercise was to determine the concentration of an unknown hydrochloric acid solution using a standardized sodium hydroxide solution. Students performed acid-base titrations and used pH measurements to identify the equivalence point. This exercise also introduced students to the use of indicators and pH meters for determining the endpoint of titrations.

Background Theory

Acid-base titration is a quantitative analytical technique used to determine the concentration of an acid or base by reacting it with a solution of known concentration. The point at which the reaction is complete is called the equivalence point.

For the reaction: $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$

The equivalence point occurs when moles of acid = moles of base, or:

$$M_a V_a = M_b V_b$$

where M is molarity and V is volume.

pH indicators change color at specific pH values. Phenolphthalein changes from colorless to pink at $\text{pH} \approx 8\text{-}10$, making it suitable for strong acid-strong base titrations.

Materials and Methods

Materials:

- Unknown HCl solution (~0.1 M)
- Standardized NaOH solution (0.1023 M)
- Phenolphthalein indicator
- pH meter (calibrated with buffer solutions)
- Buret (50 mL)
- Erlenmeyer flasks (250 mL)
- Volumetric pipet (25.00 mL)
- Distilled water

Procedure:

1. Preparation:

- Rinse buret with NaOH solution
- Fill buret with standardized NaOH solution
- Record initial buret reading

2. Titration 1 (with indicator):

- Pipet 25.00 mL of unknown HCl into Erlenmeyer flask
- Add 2-3 drops of phenolphthalein
- Titrate with NaOH, swirling constantly
- Record volume when solution turns pink (persists for 30 seconds)

3. **Titration 2-3 (with pH meter):**
- Repeat titration using pH meter to monitor pH
 - Record pH after each 1 mL addition
 - Near equivalence point, add 0.1-0.2 mL increments
 - Continue until pH stabilizes above 11
4. **Data Analysis:**
- Calculate HCl concentration from each titration
 - Construct pH vs. volume curve
 - Determine equivalence point from pH curve

Results

Raw Data

Titration	Initial Buret Reading (mL)	Final Buret Reading (mL)	Volume NaOH Used (mL)	Method
1	0.05	23.45	23.40	Indicator
2	0.02	23.52	23.50	pH meter
3	0.10	23.48	23.38	pH meter

Calculations

Using the formula: $M_{\text{HCl}} = (M_{\text{NaOH}} \times V_{\text{NaOH}}) / V_{\text{HCl}}$

Titration 1:

$$M_{\text{HCl}} = (0.1023 \text{ M} \times 23.40 \text{ mL}) / 25.00 \text{ mL} = 0.0957 \text{ M}$$

Titration 2:

$$M_{\text{HCl}} = (0.1023 \text{ M} \times 23.50 \text{ mL}) / 25.00 \text{ mL} = 0.0962 \text{ M}$$

Titration 3:

$$M_{\text{HCl}} = (0.1023 \text{ M} \times 23.38 \text{ mL}) / 25.00 \text{ mL} = 0.0956 \text{ M}$$

Average concentration: $(0.0957 + 0.0962 + 0.0956) / 3 = \mathbf{0.0958 \text{ M}}$

Standard deviation: 0.0003 M

pH Curve Analysis

The pH vs. volume curve from Titration 2 showed:

- Initial pH: 1.2 (acidic solution)
- Gradual increase to pH $\approx$ 4
- Sharp rise at equivalence point (pH jumps from $\sim$ 4 to $\sim$ 10 over 1 mL)
- Equivalence point (steepest slope): $\sim$ 23.5 mL NaOH
- Final pH: 11.5

The equivalence point determined from the pH curve matches closely with the indicator endpoint.

Discussion

The three titrations showed good reproducibility, with volume measurements agreeing within 0.15 mL (0.6% relative error). The calculated HCl concentration was 0.0958 M with a standard deviation of 0.0003 M, indicating high precision.

Indicator vs. pH Meter:

- The phenolphthalein endpoint (colorless to pink) occurred at pH $\sim$ 9, slightly after the true equivalence point (pH $\sim$ 7)

for strong acid-strong base). This is acceptable since phenolphthalein's transition range (8-10) is close to neutral.

- The pH meter method provided more precise results and allowed visualization of the entire titration curve, including the steep pH change at the equivalence point.
- Both methods gave consistent results, confirming the reliability of the analysis.

Sources of Error:

- **Buret reading:** ± 0.02 mL uncertainty in each reading leads to ± 0.04 mL total error ($\sim 0.2\%$ relative error)
- **Pipet accuracy:** ± 0.03 mL in 25.00 mL volume ($\sim 0.1\%$ relative error)
- **Indicator endpoint:** Slight overshooting possible due to delayed color change
- **pH meter calibration:** Small drift between calibrations

The combined uncertainty is approximately ± 0.001 M, giving a concentration range of 0.0958 ± 0.001 M.

Conclusion

This experiment successfully determined the concentration of an unknown HCl solution to be 0.0958 M using acid-base titration. Both the indicator method (phenolphthalein) and pH meter method produced consistent and precise results. The pH titration curve clearly showed the equivalence point through the characteristic steep pH change, demonstrating the utility of pH monitoring for endpoint detection.

The exercise reinforced key concepts in acid-base chemistry, including the stoichiometry of neutralization reactions, the use of indicators, and the interpretation of titration curves. The high precision of the results (standard deviation $< 0.5\%$) demonstrates the reliability of volumetric analysis techniques.